```
def greet(name):
    return "Hello, " + name + "!"

print(greet("Alice"))
```

Step-by-Step Guide: Writing Your First Python Program

Now that you have a basic understanding of Python syntax, let's write a simple program to reinforce these concepts. We'll create a program that asks the user for their name and age, then calculates the year they were born.

1. **Open Your IDE/Text Editor:**
   - Open your chosen IDE or text editor (e.g., Visual Studio Code).
2. **Create a New File:**
   - Create a new file and name it age_calculator.py.
3. **Write the Code:**
   - Start by asking the user for their name and age:

```
# Ask for the user's name
name = input("Enter your name: ")

# Ask for the user's age
age = int(input("Enter your age: "))
```

   - Calculate the year of birth:

```
# Import the datetime module
import datetime

# Get the current year
current_year = datetime.datetime.now().
year

# Calculate the year of birth
birth_year = current_year - age
```

   - Print the result:

```
# Display the result
  print(f"Hello, {name}! You were born
in {birth_year}.")
```

4. **Save the File:**
   - Save the file with the name age_calculator.py.
5. **Run the Program:**
   - Open your terminal and navigate to the directory where your file is saved.
   - Run the program by typing:

```
python age_calculator.py
```

6. **Test the Program:**
   - Enter your name and age when prompted.
   - Verify that the program correctly calculates and displays your year of birth.

Industry Trends and Applications

Python is widely used across various industries, and its popularity continues to grow. Here are some current trends and applications of Python:

1) Data Science and Machine Learning:

Python is the go-to language for data science and machine learning due to its powerful libraries like NumPy, pandas, and scikit-learn. Tools like TensorFlow and PyTorch make it easy to develop machine learning models.

Example: Using pandas to analyze a dataset

```
import pandas as pd

# Load a CSV file
data = pd.read_csv("data.csv")

# Display the first few rows
print(data.head())
```

2) Web Development:

Frameworks like Django and Flask make web development with Python efficient and scalable. These frameworks provide tools for building robust web applications.

Example: A simple Flask application

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
```